1.5 Division

Dividend Divisor

5771 | 135
540 | 42 ✓
=371
270
101 ✓

1101 0101 | 1001
-1001 | 10111
=10001 | 230 ten
-01001
=10000
-1001
==1111
1001
=110
6 ten

213 | 9
18 | 23
=33
27
=6

214
64
128
213

1024
2048
4096
8192

5771 -
4096
1675 -
1024
=651 -
512
139 -
128
=11

135 -
128
=7

Divisor on n bits

Dividend on n+m bits

=> Quotient on (m+1) bits

If we want the quotient on n bits => m = n-1

=> Dividend on 2n-1 bits

$5771_{ten} = 1011010001011$

$135_{ten} = 10000111$

1.5.1 Restoring division

COUNT	A	Q	M
000	0010 1101	0001 0110	1000 0111
−	1000 0111		
	1010 0110		
+	1000 0111		
	0010 1101	0001 0110	
	0101 1010	0010 1100	
001 −	1000 0111		
	1101 0011		
+	1000 0111		
	0101 1010	0010 1100	
	1011 0100	0101 1000	
010 −	1000 0111		
	0010 1101	0101 1001	
	0101 1010	1011 0010	
011 −	1000 0111		
	1101 0011		
+	1000 0111		
	0101 1010	1011 0010	
	1011 0101	0110 0100	
100 −	1000 0111		
	0010 1110	0110 0101	
	0101 1100	1100 1010	
101 −	1000 0111		
	1101 0101		
+	1000 0111		
	0101 1100	1100 1010	
	1011 1001	1001 0100	
110 −	1000 0111		
	0011 0010	1001 0101	
	0110 0101	0010 1010	
111 −	1000 0111		
	1101 1110		
+	1000 0111		
	0110 0101	0010 1010	
	101_{ten}	42_{ten}	
	Remainder	Quotient	

$n = 8$ bits

Operands

8

n subtractions

$n \leq Op \leq 2n$

$$\begin{array}{r} 37_{+} \\ 64 \\ \hline 101 \end{array}$$

13 OPS (+/−)

1.5.2 Non-restoring division

r_i

(0) → $r_i - M + M$ → $r_{i+1} = 2r_i$

(1) → $r_{i+1} = 2(r_i - M) = 2r_i - 2M$

$r_{i+1} = r_{i+1} - M$

$r_i - M$

$s = 1$ (0) → $r_{i+1} = 2r_i - 2M \xrightarrow{+M} 2r_i - M$

$s = 0$ (1) → $r_{i+1} = 2r_i - 2M \xrightarrow{-M} r_{i+1} = 2r_i - 3M$

COUNT	S	A	Q	M
000	0	0010 1101	0001 0110	1000 0111
	−	1000 0111		
	(1)	1010 0110	0001 0110	
	1	0100 1100	0010 1100	
001	+	1000 0111		
	(1)	1101 0011	0010 1100	
	1	1010 0110	0101 1000	
010	+	1000 0111		
	(0)	0010 1101	0101 1001	
	0	0101 1010	1011 0010	
011	−	1000 0111		
	(1)	1101 0011	1011 0010	
	1	1010 0111	0110 0100	
100	+	1000 0111		
	(0)	0010 1110	0110 0101	
	0	0101 1100	1100 1010	
101	−	1000 0111		
	(1)	1101 0101	1100 1010	
	1	1010 1011	1001 0100	
110	+	1000 0111		
	0	0011 0010	1001 0101	
	0	0110 0101	0010 1010	
111	−	1000 0111		
	(1)	1101 1110	0010 1010	
	+	1000 0111		
	0	0110 0101		

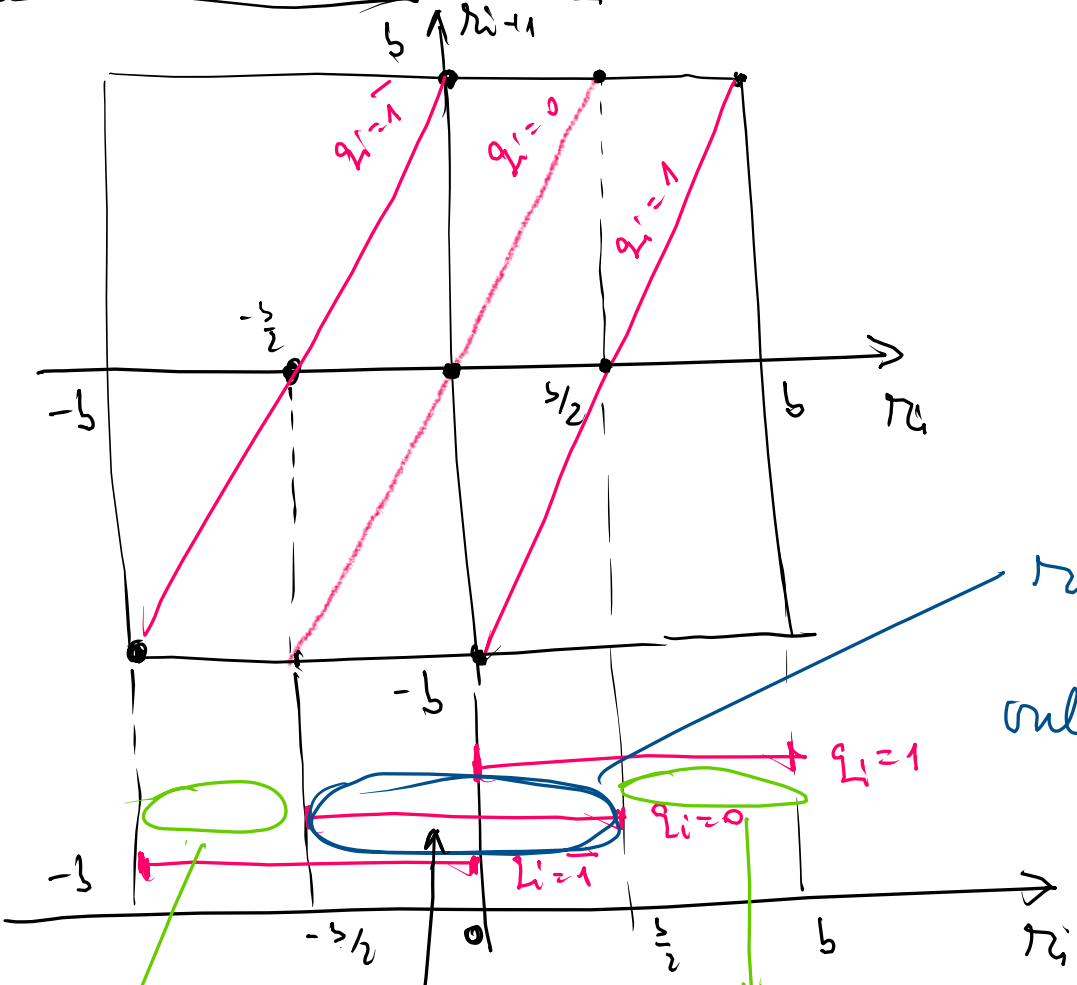
9 OPS (+/−)

1.6 Speeding division with the SRT algorithm

Radix 2 redundant digit set $\{\bar{1}, 0, 1\}$

Divisor M, b if $|r_i| < b \Rightarrow |r_{i+1}| < b$

$$\boxed{r_{i+1} = 2r_i - q_i b} \; ; \; q_i \in \{\bar{1}, 0, 1\}$$



$r_{i+1} = 2r_i$
$\Downarrow$
only shift

3 MSB of P are not equal, $P[8] = 1$

if 3 MSB of P are not equal and $P[8] = 0$

if first 3 MSB of P are equal

Example SRT radix 2

	2n+1 bits	2n bits	n bit
COUNT	P	A	B
000	000 00000 000 01101 $\xi_1 = 0$ 000 11101	1110 1101 1010 0000 0100 0000	0000 0110 1100 0000
001	$\xi_1 = 0$ 001 1011 0	1000 0000	
010	$\xi_2 = 1$ 01101101 -11000000 00010101	0000 0001	

255 -
18
237 | 6
18 | 39
= 57
54
= 3